RPoE: A Reputation-Aware Proof-of-Energy Blockchain Protocol for Electric Vehicle Energy Trading in IoEV

Koustav Kumar Mondal, *Graduate Student Member, IEEE*, Amritesh Kumar, *Member, IEEE*, and Debasis Das, *Senior Member, IEEE*

***Abstract*—Decentralized energy trading among electric vehicles (EVs) and charging stations (CSs) still suffers from high energy consumption, low throughput, and heavy consensus overheads. We present Reputation-Aware Proof-of-Energy (RPoE), a lightweight consensus that combines an energy score (ES) and a reputation score (RS); nodes with RPoE above a threshold participate in validation, and the highest-scoring node proposes the block. To sustain participation, we introduce a budget-balanced incentive with a guaranteed participation floor and a proportional share derived from concave weights of normalized RPoE and exchanged energy; tunable parameters trade fairness for efficiency. Our security analysis provides formal guarantees of double-spend resistance, Sybil resistance (trust-authority–backed identities), liveness under bounded delays, and resilience to DDoS/eclipse through eligibility gating. On a multi-node Raspberry Pi (RPi) testbed, RPoE reduces CPU workload by 47%, bandwidth by 5%, and energy consumption by 10% relative to Practical Byzantine Fault Tolerance (PBFT), Proof-of-Stake (PoS), and Proof-of-Authority (PoA), while achieving higher throughput and lower confirmation latency. The incentive remains fair and balanced. In a 10-node study with a participation floor of 20%, Jain's Fairness Index (JFI) is 0.853 (values closer to 1 indicate more fair splits) and the Gini coefficient is 0.221 (values closer to 0 indicate more equal splits), demonstrating equitable and budget-balanced rewards that avoid dominance.**

***Index Terms*—Blockchain, energy trading, energy score, vehicle to vehicle communication.**

I. Introduction

THE Internet of Electric Vehicles (IoEVs) within smart cities offers benefits such as reduced greenhouse gas emissions, cost savings, and improved grid reliability and resilience. [1]. EVs and Charging Stations (CSs) are central to IoEV success [2]. In the traditional centralized charging paradigm, energy trading relies heavily on intermediaries such as grid operators or third-party service aggregators to manage user authentication, billing, and transaction settlement. While established, this architecture suffers from inherent structural limitations. As illustrated in the left panel of Fig. 1, traditional energy trading operates on a rigid hub-and-spoke topology. In this model, all interactions between Electric Vehicles (EVs) and Charging Stations (CSs) are mediated by a central authority typically a Grid Operator or a Service Aggregator. This intermediary handles authentication, pricing, and transaction settlement, effectively creating a closed-loop system. While established, this centralized architecture imposes severe structural limitations. First, the reliance on a single central server creates a Single Point of Failure (SPoF); a server outage can paralyze the entire regional charging network. Second, the mandatory involvement of intermediaries introduces transactional opacity and informational asymmetry, often preventing users from verifying real-time pricing mechanisms [3], [4]. Finally, the aggregation of user location and consumption data in a central silo raises significant privacy concerns. To mitigate these inefficiencies, the industry is shifting toward the decentralized architecture depicted in the right panel of Fig. 1. Blockchain-based charging infrastructures leverage inherent attributes such as transparency, immutability, robust security, and resilience to facilitate direct, decentralized energy trading [5]. This approach eliminates the need for a central authority, enabling seamless transactions between electric vehicles and charging stations (E2C) as well as among charging stations (C2C) [6], [7], [8]. Consensus

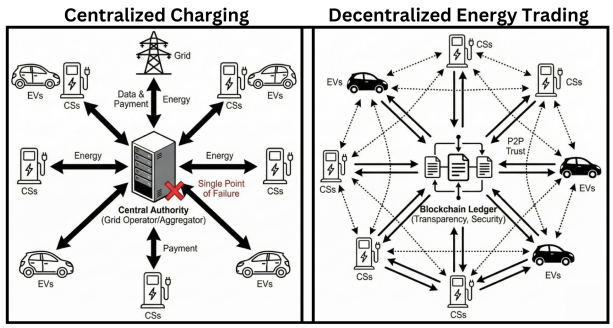


Fig. 1. Comparative architecture of EV energy trading: (Left) Traditional centralized model vs. (Right) Decentralized RPoE model.

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Koustav Kumar Mondal is with the Centre for AIoT and Applications, Indian Institute of Technology Jodhpur, Rajasthan 342030, India (e-mail: mondal.4@iitj.ac.in).

Amritesh Kumar is with the Centre for AIoT and Applications, Indian Institute of Technology Jodhpur, Rajasthan 342030, India, and also with the School of Computer Science Engineering and Technology (SCSET), Bennett University, Greater Noida 201310, India (e-mail: amritesh@bennett.edu.in).

Debasis Das is with the Department of Computer Science and Engineering, Indian Institute of Technology Jodhpur, Rajasthan 342030, India (e-mail: debasis@iitj.ac.in).

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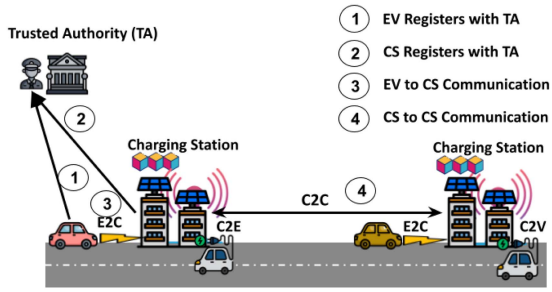


Fig. 2. Architectural Overview of the Proposed RPoE Scheme.

algorithms - PoW [9], [10], PoS [11], PoA [12], PBFT [13], and PoR [14], [15] - provide integrity and transparency, but in IoEV energy trading they often yield low throughput and high energy/communication overheads, limiting city-scale deployment [16], [17], [18]. Key gaps include (i) poor scalability and rising latency from PBFT-style messaging, (ii) centralization risk and weak energy awareness in stake/authority schemes, (iii) incentives loosely coupled to actually delivered/consumed energy (weak participation/fairness), and (iv) on-chain settlement costs and metadata exposure that strain edge devices and privacy.To address these limitations, we introduce a lightweight and energy-aware consensus mechanism with an incentive layer called Reputation-Aware Proof-of-Energy (RPoE). Each node's scalar fitness $RPoE_i$ (9) combines energy and reputation; nodes with $RPoE_i \geq \theta$ are eligible, and the proposer is the eligible node with the maximum $RPoE$. Per epoch, rewards are allocated by a budget-balanced floor+proportional rule (20) using concave contribution weights (19); the floor β guarantees a non-zero minimum, while $\gamma, \delta \leq 1$ induce diminishing returns with clipping caps (R^*, Q^*) to bound outliers (Section III-F). This couples validator selection and incentives to measured behavior, curbs dominance, and sustains participation without PBFT-style messaging overheads.

A. Motivation

The drivers of electric vehicles (EVs) encounter difficulties when undertaking long-distance travel due to battery limitations and the need for accessible charging stations (CS) for energy trading [38], [39]. Traditional energy trading between nodes, such as EVs and CSs, is plagued by challenges related to transparency, ownership verification, energy consumption measurement, and security risks [3]. Blockchain technology emerges as a viable solution to address these hurdles by providing a secure and efficient platform for decentralized energy sharing [40]. However, the existing consensus mechanisms employed by blockchain, such as Proof of Work (PoW) [9], [10] and Proof of Stake (PoS) [11], prove to be inefficient for energy exchange due to their high computational and energy requirements. Alternatives like Proof of Authority (PoA) [12] and off-chain transactions possess their own limitations, including trust dependency and compromised security and transparency. These challenges motivate RPoE, a consensus mechanism that combines a node's reputation score (RS) and energy score (ES) into a unified fitness; nodes with $RPoE \geq \theta$ participate in validation and the leader is chosen by $\arg\max RPoE$. Complementing selection, we employ a budget-balanced, fairness-tunable incentive scheme that splits each epoch's reward into a participation floor (guaranteeing a non-zero share to every eligible node) and a proportional share allocated via concave weights derived from clipped-normalized reputation and energy factors. Together, these design choices target an efficient, secure, and transparent energy-trading substrate for IoEVs, promoting broad participation, reliability, and the growth of a sustainable energy ecosystem.

B. Contributions

The proposed scheme advances blockchain-based energy trading for IoEV along both consensus and incentivization axes. Our key contributions are:

- *RPoE consensus with calibrated energy emphasis.* We introduce Reputation-Aware Proof-of-Energy (RPoE), where validator fitness is a bilinear combination of energy and reputation scores, $RPoE = EF \cdot ES + RF \cdot RS$. The weights (EF, RF) are obtained by solving linear constraints that prioritize energy efficiency while remaining balanced (14). Nodes with $RPoE \geq \theta$ are eligible; leaders are selected by $\arg\max RPoE$.
- *Fair and budget-balanced incentive rule.* We design a per-epoch incentive mechanism (20) that (i) exactly conserves the budget, (ii) guarantees a non-zero floor to every eligible node via parameter $\beta \in [0, 1)$, and (iii) allocates the residual budget proportionally to concave contribution weights $w_i = (\hat{R}_i)^\gamma (\hat{Q}_i)^\delta$ with clipped-normalized factors $(\hat{R}_i, \hat{Q}_i) \in [0, 1]$. We prove key properties: budget balance and individual rationality; monotonicity with diminishing returns for $\gamma, \delta < 1$; symmetry among equals; and proportional fairness of the variable allocation via a weighted log-utility maximization. An optional exponential smoother stabilizes rewards over time.
- *Fairness-tunable incentive with formal metrics and empirical gains.* A budget-balanced floor + proportional rule exposes three knobs—participation floor β, concavity (γ, δ), and clipping caps (R^*, Q^*)—to trade equity/efficiency and bound outliers. Fairness is measured via Jain's index and Gini (Lorenz curves; Figs. 7–9, Table V). In a 10-node study with $(\gamma, \delta) = (0.6, 0.8)$, $\beta = 0.2$ yields $J = 0.853$ and $G = 0.221$; increasing β from 0 to 0.4 raises J from 0.788 to 0.912. The end-to-end pipeline (Fig. 3) and Raspberry Pi testbed show CPU -47%, bandwidth -5%, and energy -10% vs. baselines, with low latency and favorable throughput scaling.
- *Security arguments.* We present concise lemmas for double-spend resistance, Sybil resistance (TA-backed identities with verifiable metrics), liveness under bounded delays, and resilience against DDoS/eclipse via eligibility gating on high-$RPoE$ nodes.

II. Brief Literature Review

Blockchain-based energy trading (BET) aims to enable secure, auditable exchanges among electric vehicles (EVs) and charging stations in a decentralized manner. Classical consensus mechanisms offer different trade-offs. Proof-of-Work (PoW) provides robust Sybil resistance but imposes excessive energy expenditure and specialized hardware demands, which are misaligned with sustainability goals in IoEV ecosystems [19].

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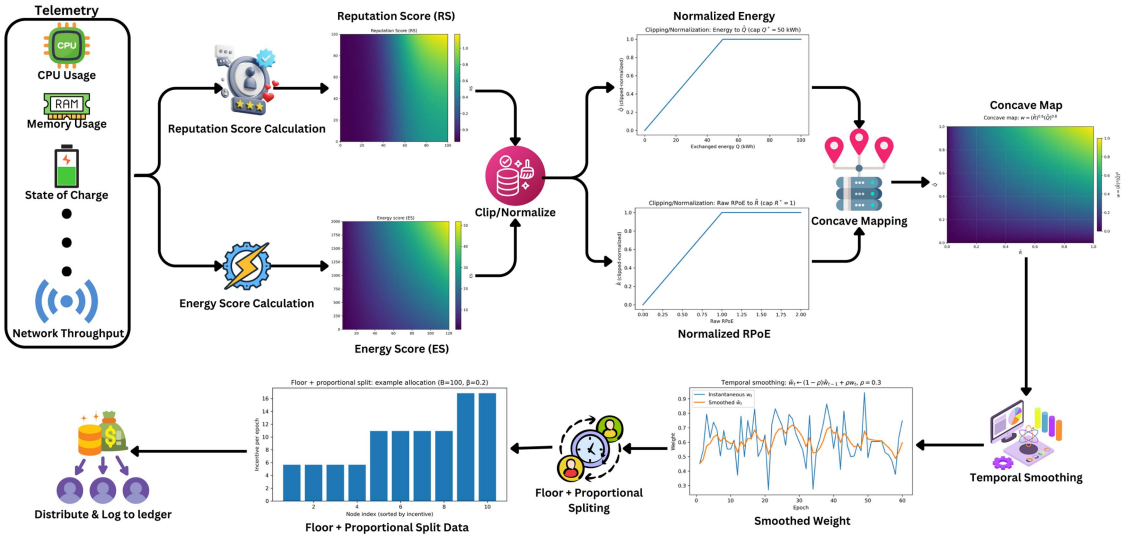


Fig. 3. Proposed Incentive Calculation Pipeline.

Proof-of-Stake and its delegated variants (PoS/DPoS) reduce computational burden but inherit centralization risks and exhibit a weak coupling to physical energy efficiency [20]. In permissioned settings, PBFT remains popular; however, its $O(n^2)$ message complexity and view-change overheads constrain scalability at city-scale deployments [21].

Recent studies have reinforced these classical observations. A hybrid blockchain consensus combining PoW and PoS was shown to enhance throughput and scalability in EV energy markets while maintaining decentralized control [29]. Similarly, comparative benchmarking of PBFT, Tendermint, and PoET for microgrid-scale BET confirms the persistent trade-offs between security, latency, and communication overheads [30]. Updated surveys echo these challenges, emphasizing scalability and fairness as ongoing issues in energy-aware blockchain systems [34].

To better suit energy domains, several works tailor consensus or system design to EV/V2G contexts. PDPoS adapts delegated stake for BET [22]; PoR and its variants (PPoR) incorporate reputation into selection [3], [23]; BAC advances consortium-chain performance via BFT-style coordination [25]; and reputation-based D-BFT exploits trust signals in permissioned networks [26]. Extending this trend, newer frameworks integrate explicit reputation management to strengthen trust in distributed markets: Zhang et al. propose a reputation-aware secure transactive energy market with blockchain coordination [31]; Zhao et al. design a reputation-based consensus for distributed urban energy trading that improves transparency and throughput [32]; and Mo et al. develop a lightweight reputation-aware BET framework optimized for distributed IoEVs [33].

Very recently, advanced privacy-preserving reputation schemes have emerged to address overhead and role-specific attacks. Liu et al. proposed PPRU [35], a cloud-assisted framework utilizing Paillier homomorphic encryption to offload reputation aggregation to a Cloud Service Provider (CSP), thereby reducing the Trusted Authority's computation load by over 80%. Similarly, Sun et al. introduced PPDR [36] for vehicle platoons, which employs a dual-reputation mechanism (separating leading and following vehicle scores) to mitigate single-reputation attacks. While these works represent the -of-the-art in privacy and trust management, they primarily function as overlay management systems relying on semi-trusted entities (Cloud/TA). Crucially, they lack the physical energy coupling central to RPoE; they do not mathematically bind a node's voting power to its verifiable energy efficiency (SoC, charging behavior), nor do they provide a budget-balanced tokenomic incentive model for decentralized trading.

However, a critical gap remains in how these hybrid schemes utilize reputation. In most recent variants (e.g., [31], [33]), the reputation score serves primarily as a security filter to exclude malicious actors, while voting power often remains tied to stake or static authority. These designs typically decouple the consensus mechanism from the physical layer—meaning a node's voting influence is not mathematically tied to its real-time energy efficiency or service quality.

A complementary line of research implements market logic via smart contracts on public chains (e.g., pricing, settlement, and strategy stabilization) [2], [6], [27], [28]. While these approaches add incentive integration at the application layer, they do not modify consensus properties, incur on-chain fees, and can raise metadata exposure concerns. Recent energy-market smart contracts extend these ideas through cross-chain trading and adaptive incentive policies but continue to suffer from high gas costs and on-chain latency [34]. Comprehensive surveys concur that current BET solutions still face gaps in throughput, messaging overhead, and energy/utility efficiency [16], [17], [37].

Against this backdrop, the proposed RPoE (bottom row of Table I) addresses this decoupling by introducing a bilinear fitness function that strictly binds validator eligibility to verifiable energy telemetry (ES) and reputation (RS). Unlike hybrid schemes that optimize for generic throughput, RPoE optimizes for *utility*, ensuring that the nodes maintaining the ledger are those actively contributing to the energy network's efficiency. Furthermore, it integrates a budget-balanced floor with fairness knobs (β, γ, δ) for incentive allocation, incorporating dominance mitigation via clipping/smoothing, and reducing coordination costs by confining consensus to high-fitness candidates in a permissioned setting.

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TABLE I
BROADER COMPARISON ACROSS EXISTING APPROACHES AND THE PROPOSED RPOE. BET = BLOCKCHAIN-BASED ENERGY TRADING. "PERM." = PERMISSIONED. FAIRNESS KNOBS (β, γ, δ) AND DOMINANCE MITIGATION (CLIPPING/SMOOTHING) ARE EXPLICIT IN RPOE. COMMUNICATION OVERHEAD IS QUALITATIVE. †MARKET-LEVEL INCENTIVES VIA SMART CONTRACTS, NOT CONSENSUS-COUPLED.

Approaches	BET Focus	Perm. Model	Consensus / Scheme	RA	ES	II	BBF	FK	DM	CO	E/IE
PoW [19]	✗	Public	PoW	✗	✗	✗	✗	✗	✗	High	✗
PoS/DPoS (surveyed) [20]	✗	Public/Perm.	PoS/DPoS	✗	✗	✗	✗	✗	✗	Med	✗
PBFT [21]	✗	Permissioned	PBFT	✗	✗	✗	✗	✗	✗	High $(O(n^2))$	✗
PDPoS [22]	✓	Consortium	DPoS-variant	✗	✗	✗	✗	✗	✗	Med	✗
PoR for BET [23]	✓	Consortium	PoR	✗	✗	✗	✗	✗	✗	Med	✗
PPoR [3]	✓	Consortium	PoR-variant	✗	✗	✗	✗	✗	✗	Med	✗
RBCET [24]	✓	Permissioned	RBCET	✓	✗	✓	✗	✗	✗	Med	✓
BAC (consortium) [25]	✓	Permissioned	BFT-variant	✗	✗	✗	✗	✗	✗	Med	✗
Reputation-based D-BFT [26]	✗	Permissioned	D-BFT	✓	✗	✗	✗	✗	✗	Med	✗
EC[6], [2], [27], [28]	✓	Public	PoW/PoS (underlying)	✗	✗	✓†	✗	✗	✗	Med–High (gas)	✗
Hybrid BET (PoW-PoS) [29]	✓	Consortium	Hybrid PoW/PoS	✗	✓	✗	✗	✗	✗	Med	✓
Consensus Benchmarking (PBFT/Tendermint/PoET) [30]	✓	Consortium	PBFT/Tendermint/PoET	✗	✗	✗	✗	✗	✗	Med–High	✓
Reputation-Aware BET [31]	✓	Permissioned	Reputation-based	✓	✗	✓	✗	✗	✗	Med	✓
Reputation-based Consensus for Urban Energy [32]	✓	Permissioned	Reputation-Consensus	✓	✓	✓	✗	✗	✗	Med	✓
Lightweight Reputation-Aware BET [33]	✓	Permissioned	Lightweight BFT-variant	✓	✓	✓	✗	✗	✗	Low–Med	✓
Blockchain-Smart Grid Survey [34]	–	–	–	✗	✗	✗	✗	✗	✗	–	–
PPRU [35]	Gen. VANET	Cloud-Asst.	N/A (Centralized)	✓	✗	✗	✗	✗	✓	Low	✓
PPDR [36]	Platoon	Cloud-Asst.	N/A (Centralized)	✓(Dual)	✗	✗	✗	✗	✓	Med	✓
Surveys [16], [17], [37]	–	–	–	✗	✗	✗	✗	✗	✗	–	–
Proposed RPoE	✓	Permissioned	Reputation+Energy (RPoE)	✓	✓	✓	✓	✓(β, γ, δ)	✓(clipping/smoothing)	Low–Med	✓

EC: Ethereum-based contracts, ES: Energy-Aware Selection, RA: Reputation Assess, BBF: Budget-Balanced Floor, II: Incentive Integration, DM: Dominance Mitigation, CO: Communication Overhead, E/IE: Edge/IoEV Efficiency.

III. PROPOSED METHODOLOGY

We architect a permissioned blockchain overlay for IoEV in which Electric Vehicles (EVs) and Charging Stations (CSs) transact energy through application smart contracts, while validator participation and block proposing are governed by the Reputation-Aware Proof-of-Energy (RPoE) mechanism. Fig. 2 summarizes the components and message flows.

- *Identity and admission (Fig. 2 ① & ②).* Each node onboards via a Trust Authority (TA) that issues a verifiable identity (e.g., X.509/DID) binding the node to its metering and telemetry sources. The TA anchors the public key and role metadata on-chain, enabling authenticated messages and Sybil resistance.
- *Telemetry anchoring and inputs to scoring.* At fixed intervals, each node signs and submits a compact telemetry anchor (e.g., a Merkle root) that commits to recent operational traces and metering summaries. These anchors provide the deterministic inputs for the reputation score RS (CPU, memory, of charge; (5)) and the energy score ES (driving/environmental factors), which are combined into $RPoE$ as per (9) with weights calibrated by (14). To control on-chain load, raw traces may remain off-chain while their cryptographic digests are recorded on-chain.
- *Epoching, eligibility, and committee formation.* Time is partitioned into epochs. At the start of epoch t, all nodes deterministically compute $\{RS_i, ES_i, RPoE_i\}$ from the latest anchors and derive the eligible set $\mathcal{S}_t = \{i : RPoE_i \geq \theta\}$ using the protocol threshold θ (configurable; median-adaptive in our experiments). The hash of $\mathcal{S}_t$ (and the vector of scores) is recorded on-chain for auditability.
- *Leader selection and block proposal.* The block proposer for the next slot is chosen as $i^\star = \arg\max_{i \in \mathcal{S}_t} RPoE_i$ with a deterministic tie-breaker (e.g., node ID hash). The proposal contains: application transactions (E2C, C2C), the epoch tag, the proposer identity, the digest of the epoch's $RPoE$ snapshot, and the incentive allocation digest for this epoch. This narrows consensus traffic to high-quality nodes and reduces coordination overhead versus all-to-all BFT messaging.
- *Validation and commit.* Validators in $\mathcal{S}_t$ verify (i) transaction correctness and signatures, (ii) the proposer's score evidence against the published telemetry anchors, and (iii) the incentive allocation recomputed locally. Upon reaching a quorum (e.g., $2f+1$ eligible signatures in a fault-tolerance setting), the block is committed and finalized.
- *Incentive settlement on-chain.* Per-epoch budget B_t is distributed by a settlement contract implementing (20), using contribution weights w_i from (19) (with clipping caps and optional smoothing as specified in Section III-F). The contract mints/transfers credits to eligible nodes, ensuring budget balance and a guaranteed participation floor. Receipts are written to the ledger for audit.
- *Application workflow (Fig. 2 ③ & ④).* EV–CS (E2C) and CS–CS (C2C) trades are executed via application contracts that handle offers, metering proofs, clearing, and settlement. These contracts are agnostic to RPoE; they consume the same finalized ledger, while RPoE governs who proposes and validates blocks.

Deterministic score computation from on-chain telemetry anchors aligns validator selection with measured operational and energy behavior; eligibility gating together with single–proposer selection reduces CPU load, bandwidth, and confirmation latency on resource-constrained EV/CS hardware; and on-chain settlement of the floor + proportional incentive in (20) ties rewards to normalized $RPoE$ and exchanged energy, offering formal fairness controls (Section III-F) while preserving liveness and auditability. With the blockchain coupling and incentive plumbing established, we next develop the Mathematical Modeling of Node Reputation Parameters, which provides the inputs for RS and, in turn, for $RPoE$.

A. Mathematical Modeling of Node Reputation Parameters

To accurately capture the complex interdependencies among node parameters while reflecting real-world variabilities, we propose augmented formulations for reputation score estimation. These formulations introduce additional corrective and nonlinear terms to encapsulate secondary effects and stochastic variations observed in empirical data.

a) Model for CPU and memory usage: Rather than a simple linear dependence, we incorporate a logarithmic correction term and an additive noise factor to account for minor

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fluctuations due to operational overhead:

$$M = k\,C + b + \alpha \ln(1 + C) + \epsilon_M, \tag{1}$$

where M represents memory usage, C denotes CPU usage, $k > 0$ and $b \in \mathbb{R}$ are baseline parameters, and $\alpha > 0$ scales the logarithmic correction capturing diminishing incremental memory demands at higher CPU loads. The term $\epsilon_M \sim \mathcal{N}(0, \sigma_M^2)$ models inherent system noise. This formulation was derived based on regression analysis from system performance data, revealing that small nonlinearities improve model fidelity.

b) Model for SoH and SoC: Battery performance characteristics often display nonlinear degradation behavior. We augment the quadratic model with an exponential decay component to capture accelerated battery wear at lower charge levels:

$$\text{SoH} = a\,\text{SoC}^2 + b\,\text{SoC} + c + \gamma\,\exp\left(-\delta(1 - \text{SoC})\right) + \epsilon_S, \tag{2}$$

where SoH is the of health, and SoC is the of charge. Here, a, b, and c are quadratic coefficients, $\gamma > 0$ and $\delta > 0$ parameterize the exponential term accounting for rapid degradation as SoC decreases, and $\epsilon_S \sim \mathcal{N}(0, \sigma_S^2)$ represents measurement noise. This model was developed following empirical battery discharge analyses which indicated that a pure quadratic model underestimates degradation phenomena under low charge conditions.

c) Model for network throughput vs. CPU usage: Network throughput is influenced by CPU usage in a sigmoidal manner due to resource saturation [24], [41], [42]; however, additional intermediate dynamics necessitate a composite model. We propose:

$$\text{NT} = \frac{K}{1 + \exp\left(-p(C - q)\right)} + \beta\,\tanh\left(r(C - s)\right) + \epsilon_{NT}, \tag{3}$$

where NT is network throughput, C is CPU usage, and K is the maximum achievable throughput. The first term is the standard logistic function capturing the initial rapid increase and eventual saturation, with $p > 0$ controlling the growth rate and q the inflection point. The additive term $\beta\,\tanh(r(C - s))$ (with $\beta, r > 0$ and $s \in \mathbb{R}$) models intermediate oscillatory behavior observed in systems approaching saturation. Finally, $\epsilon_{NT} \sim \mathcal{N}(0, \sigma_{NT}^2)$ accounts for stochastic variations. This composite model was inspired by signal processing techniques where hybrid functions improve fit accuracy in nonlinear regimes.

Each equation has been derived through statistical analysis and curve-fitting of observed data, ensuring that they not only align with theoretical expectations but also robustly capture the variability inherent in real-world operational conditions. The incorporation of additional nonlinear components and noise terms reflects our research's commitment to achieving a more nuanced

B. Reputation Score Calculation

The reputation score (RS) quantifies a node's trustworthiness using its key operational parameters: CPU usage (C), memory usage (M), and of charge (SoC). Our benchmarking validates the use of a Decision Tree Regressor (DTR) for RS prediction by optimizing feature splits based on maximum information gain:

$$IG = ET_P - \sum_{j=1}^{k} \frac{n_j}{N}\,ET_{C_j}, \quad ET = -\sum_{i=1}^{C} p_i \log_2(p_i), \tag{4}$$

where ET_P is the parent entropy, ET_{C_j} that of the j-th child, with n_j samples and N total samples. To robustly capture nonlinear interactions, we define the RS as:

$$RS = \frac{\theta_1 M + \theta_2 + \theta_3\,\text{SoC}^2 + \theta_4\,\text{SoC} + \theta_5}{1 + \exp\left(-\theta_6(C - \theta_7)\right)} + \theta_8\,\tanh\left(\theta_9(C - \theta_{10})\right) \tag{5}$$

where θ_i $(i = 1, \ldots, 10)$ are empirically optimized parameters. The quadratic term in SoC and linear term in M capture primary effects, while the logistic and hyperbolic tangent functions model CPU saturation and intermediate dynamics. This concise formulation provides a mathematically rigorous and technically robust measure of node reputation in decentralized energy trading.

C. Mathematical Modeling of Energy Parameters and Energy Score Calculation

We define the energy consumed (E_{used}) in kWh as $E_{used} = \frac{\text{DT}\cdot\text{EC}\cdot\eta}{1000}$, where DT is the distance traveled (miles), EC is the elevation change (ft), η is the efficiency of the vehicle (kWh/mile). The charging time (CT) (hours) is given by $CT = \frac{E_{used}}{CR_{\max}}$, with $CR_{\max}$ representing the maximum charging rate (kW). Temperature (T) and elevation (EV) are modeled as normally distributed random variables:

$$f_T(T) = \frac{1}{\sqrt{2\pi\,\sigma_T^2}} \exp\left(-\frac{(T - \mu_T)^2}{2\,\sigma_T^2}\right), \tag{6}$$

$$f_{EV}(EV) = \frac{1}{\sqrt{2\pi\,\sigma_{EV}^2}} \exp\left(-\frac{(EV - \mu_{EV})^2}{2\,\sigma_{EV}^2}\right), \tag{7}$$

where μ_T, σ_T and μ_{EV}, σ_{EV} are the means and standard deviations for temperature and elevation, respectively. The energy score (ES) is defined as

$$ES = \frac{BC \cdot \text{DT}}{E_{used}} \cdot \left(1 + \frac{T}{100}\right) \cdot \left(1 + \frac{\text{EC}}{1000}\right) \cdot \frac{CT}{10}, \tag{8}$$

where BC is the battery capacity. In this formulation, $\frac{BC\cdot\text{DT}}{E_{used}}$ captures the efficiency by normalizing the distance traveled with respect to energy consumption, $(1 + \frac{T}{100})$ and $(1 + \frac{\text{EC}}{1000})$ serve as multiplicative adjustment factors that account for environmental influences (temperature and elevation effects), and $\frac{CT}{10}$ normalizes the charging time relative to a standard charging rate. This rmultivariate approach mathematically encapsulates the interplay between vehicle performance metrics and environmental conditions, yielding a comprehensive energy score for blockchain-based IoEV systems.

D. RPoE Value Calculation

We define RPoE value as a bilinear combination of energy and reputation metrics:

$$RPoE = EF \cdot ES + RF \cdot RS, \tag{9}$$

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where ES and RS denote the energy and reputation scores, respectively, and EF and RF are their associated weighting factors. To calibrate these factors such that energy performance is emphasized, we impose the linear constraints

$$EF\, ES_{\max} + RF\, RS_{\min} = 0.7, \tag{10}$$

$$EF\, ES_{\min} + RF\, RS_{\max} = 0.3, \tag{11}$$

where $ES_{\max}$, $ES_{\min}$ are the maximum and minimum energy scores, and $RS_{\max}$, $RS_{\min}$ are the maximum and minimum reputation scores within the network. Expressing (10) and (11) in matrix form,

$$\begin{pmatrix} ES_{\max} & RS_{\min} \\ ES_{\min} & RS_{\max} \end{pmatrix} \begin{pmatrix} EF \\ RF \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}, \tag{12}$$

we denote the coefficient matrix by A and its determinant by $\Delta = ES_{\max}\, RS_{\max} - ES_{\min}\, RS_{\min}$. Assuming $\Delta \neq 0$, the unique solution is given by

$$\begin{pmatrix} EF \\ RF \end{pmatrix} = A^{-1} \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}, \tag{13}$$

which yields

$$EF = \frac{0.7\, RS_{\max} - 0.3\, RS_{\min}}{\Delta},$$

$$RF = \frac{0.3\, ES_{\max} - 0.7\, ES_{\min}}{\Delta}. \tag{14}$$

This formulation ensures a higher weighting of the energy factor. Hence, nodes with superior energy efficiency (high ES) are preferentially selected in the consensus process, while also considering their reputation (RS). Nodes with $RPoE \geq \theta$ (with θ a preset threshold) are eligible to participate in energy trading. This mathematically derivation via matrix inversion not only balances the two metrics but also provides a systematic approach to parameter calibration under defined performance criteria. Consistent with the adaptive framework established in Section III.E, the calibration of EF and RF is performed per epoch. The boundary values $ES_{\max}$ and $RS_{\min}$ are updated based on the currently active set $\mathcal{N}_{active}$. This ensures the weighting factors automatically adjust to shifts in network traffic and energy patterns.

E. Dynamic Thresholding and Consensus Liveness

To address the high mobility and dynamic churn inherent in IoEV networks, RPoE departs from static validator sets. We introduce an adaptive eligibility threshold and a deterministic fallback mechanism to ensure liveness during node failures.

1) Adaptive Eligibility Threshold (θ_t): Rather than a fixed, hard-coded value, the eligibility threshold θ adapts to the network's performance distribution per epoch. For an epoch t, let $\mathbf{RPoE}_t$ be the set of scores for all active nodes. The threshold θ_t is defined as the median of this distribution:

$$\theta_t = \text{Median}(\{RPoE_i \mid i \in \mathcal{N}_{active}\}). \tag{15}$$

Consequently, the eligible validator set becomes $\mathcal{S}_t = \{i : RPoE_i \geq \theta_t\}$. This ensures that the consensus committee always comprises the upper 50% of performing nodes, maintaining high validation quality even as the total population fluctuates due to vehicle entry and exit.

2) Proposer Failure and View Change: While the primary proposer is identified as $i^\star = \arg\max_{i \in \mathcal{S}_t} RPoE_i$, mobile nodes may experience abrupt disconnections (crash faults). To preserve liveness, we define a deterministic View Change protocol:

1) *Timeout:* If validators do not receive a valid block proposal from $i^\star$ within a bounded time Δ_{prop}, the node is marked as unresponsive.
2) *Deterministic Fallback:* The proposal right rotates to the node with the next-highest RPoE score:
$$i^{**} = \arg \max_{j \in \mathcal{S}_t \setminus \{i^\star\}} RPoE_j. \tag{16}$$
3) *Commit:* This rotation continues in descending order of RPoE scores until a valid block is proposed and committed by the quorum $(2f + 1)$.

This mechanism ensures that the ledger continues to progress even if the highest-reputation nodes depart the network unexpectedly.

F. Incentive Calculation

We distribute a fixed per-epoch reward budget $B_t > 0$ (e.g., tokens or energy credits) among the eligible participants $\mathcal{S}_t = \{i : RPoE_i \geq \theta\}$, where $|\mathcal{S}_t| = n_t$. To ensure fairness and robustness, we employ (i) concave contribution weights, (ii) a non-zero floor so every eligible node is rewarded, and (iii) a budget-balanced allocation with proportional-fairness guarantees as shown in the Fig. 3.

1) Normalized Contributions: Let $R^* > 0$ and $Q^* > 0$ be reference caps (e.g., high quantiles or protocol maxima). Define clipped-normalized factors

$$\hat{R}_i \triangleq \frac{\min\{RPoE_i, R^*\}}{R^*} \in [0, 1], \tag{17}$$

$$\hat{Q}_i \triangleq \frac{\min\{Q_i, Q^*\}}{Q^*} \in [0, 1]. \tag{18}$$

With concavity parameters $\gamma, \delta \in (0, 1]$, the weight of node i in epoch t is

$$w_i^{(t)} = (\hat{R}_i)^\gamma\, (\hat{Q}_i)^\delta. \tag{19}$$

Concavity ($\gamma, \delta \leq 1$) imposes diminishing returns, curbing dominance by extreme values and thus improving fairness.

2) Fair Split With Participation Floor: Let $\beta \in [0, 1)$ control the guaranteed minimum. The incentive to node $i \in \mathcal{S}_t$ is

$$I_i^{(t)} = B_t \left(\frac{\beta}{n_t} + (1 - \beta)\, \frac{w_i^{(t)}}{\sum_{j \in \mathcal{S}_t} w_j^{(t)}} \right). \tag{20}$$

Thus every eligible node receives at least $B_t \beta / n_t$, while the remaining budget $B_t(1 - \beta)$ is shared proportionally to the scientifically grounded weights (19).

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3) Temporal Smoothing: To reward sustained good behavior and reduce volatility, weights can be exponentially averaged:

$$\tilde{w}_i^{(t)} \;=\; (1-\rho)\,\tilde{w}_i^{(t-1)} \;+\; \rho\, w_i^{(t)}, \quad \rho \in (0,1], \tag{21}$$

and $\tilde{w}_i^{(t)}$ substituted for $w_i^{(t)}$ in (20).

4) Fairness and Efficiency Guarantees: We recall the per-epoch allocation ((20)) over the eligible set $\mathcal{S}_t = \{i : RPoE_i \geq \theta\}$ with size $n_t = |\mathcal{S}_t|$, budget $B_t > 0$, floor parameter $\beta \in [0,1)$, and weights $w_i^{(t)}$ defined in (19). For readability we drop the superscript (t) when unambiguous.

Lemma 1 (Budget balance and individual rationality): $\sum_{i\in\mathcal{S}_t} I_i = B_t$ and $I_i \geq B_t\,\beta/n_t > 0$ for all $i \in \mathcal{S}_t$.

Proof: Summing (20) over i gives $\sum_i I_i = B_t(\beta + (1-\beta)\frac{\sum_i w_i}{\sum_j w_j}) = B_t$. Since $w_i \geq 0$ and $\sum_j w_j > 0$ whenever $\mathcal{S}_t \neq \varnothing$, each I_i satisfies $I_i \geq B_t\,\beta/n_t$, establishing a strictly positive minimum reward per eligible node.

Lemma 2 (Rank preservation, symmetry, and no justified envy among equals): If $w_i = w_k$ then $I_i = I_k$ (symmetry). If $w_i > w_k$ then $I_i > I_k$ (rank preservation). Hence, there is no justified envy among equals.

Proof: From (20), the floor term is identical for all eligibles. The variable term is $B_t(1-\beta)\,w_i/\sum_j w_j$, which is strictly increasing in w_i holding $\{w_j\}_{j\neq i}$ fixed. Equal (resp. larger) weights therefore induce equal (resp. larger) allocations.

Lemma 3 (Monotonicity and diminishing returns in own contribution): Fix $\{w_j\}_{j\neq i}$. Then I_i is strictly increasing in $\hat{R}_i$ and in $\hat{Q}_i$. Moreover, for $\gamma, \delta \in (0,1]$, the marginal gains $\partial I_i/\partial \hat{R}_i$ and $\partial I_i/\partial \hat{Q}_i$ are nonincreasing in $\hat{R}_i, \hat{Q}_i$ (diminishing returns).

Proof: Write $S = \sum_j w_j$ and $w_i = (\hat{R}_i)^\gamma (\hat{Q}_i)^\delta$. Using the quotient rule on (20) (floor term is constant),

$$\frac{\partial I_i}{\partial \hat{R}_i} = B_t(1-\beta)\,\frac{\partial w_i}{\partial \hat{R}_i}\,\frac{S - w_i}{S^2}, \quad \frac{\partial w_i}{\partial \hat{R}_i} = \gamma\,\hat{R}_i^{\gamma-1}\hat{Q}_i^{\delta} \geq 0.$$

Since $S - w_i = \sum_{j\neq i} w_j \geq 0$ and $S > 0$, we have $\partial I_i/\partial \hat{R}_i \geq 0$. An analogous expression holds for $\partial I_i/\partial \hat{Q}_i \geq 0$. For diminishing returns, differentiate again: $\frac{\partial^2 w_i}{\partial \hat{R}_i^2} = \gamma(\gamma-1)\,\hat{R}_i^{\gamma-2}\hat{Q}_i^{\delta} \leq 0$ for $\gamma \in (0,1]$, and note that $\partial I_i/\partial \hat{R}_i$ is a product of a nonnegative term and a factor that (i) inherits the concavity of w_i in $\hat{R}_i$ and (ii) is further attenuated by the increasing denominator S. A direct but routine calculation shows $\partial^2 I_i/\partial \hat{R}_i^2 \leq 0$ whenever $\sum_{j\neq i} w_j > 0$. The argument for $\hat{Q}_i$ is identical, using $\delta \in (0,1]$.

Lemma 4 (Weighted proportional fairness of the variable part; uniqueness and Pareto efficiency): For $\beta = 0$, the allocation $I_i = B_t\,\frac{w_i}{\sum_j w_j}$ is the unique maximizer of $\max_{\{x_i\geq 0\}} \sum_{i\in\mathcal{S}_t} w_i \log x_i$ such that $\sum_{i\in\mathcal{S}_t} x_i \leq B_t$. For $\beta > 0$, the same holds on the residual budget $B_t(1-\beta)$ with decision variables $\{x_i\}$ representing only the variable portion. In all cases, the variable allocation is Pareto efficient.

Proof: The objective is strictly concave on $\mathbb{R}^{n_t}_{>0}$ (sum of concave log), and the feasible set is compact and convex; hence a unique maximizer exists. Form the Lagrangian $\mathcal{L}(x,\lambda,\nu) = \sum_i w_i \log x_i - \lambda(\sum_i x_i - B) - \sum_i \nu_i x_i$ with $B = B_t$ (or $B_t(1-\beta)$). KKT conditions yield, for each i, $\partial\mathcal{L}/\partial x_i = w_i/x_i - \lambda - \nu_i = 0$, with $\nu_i \geq 0$, $x_i \geq 0$, and complementarity $\nu_i x_i = 0$. At optimum, $x_i > 0$ (weights $w_i \geq 0$ and finite budget), so $\nu_i = 0$ and $x_i = w_i/\lambda$. Enforcing the budget constraint at equality (strict concavity) gives $\sum_i w_i/\lambda = B \Rightarrow \lambda = \frac{\sum_j w_j}{B}$ and thus $x_i = B\,\frac{w_i}{\sum_j w_j}$. Uniqueness follows from strict concavity. Proportional-fair allocations are known to be Pareto efficient; here, any feasible increase in some x_i requires a decrease in others' allocations that reduces $\sum_i w_i \log x_i$, contradicting optimality.

Lemma 5 (Robustness and bounded influence): With clipping caps $R^*, Q^* > 0$, we have $0 \leq w_i \leq 1$ and $0 < \sum_j w_j \leq n_t$. Hence the share $\frac{w_i}{\sum_j w_j} \in [0,1]$ is well–defined, and a single node's maximum influence on the variable budget is bounded by 1 (achieved only if $w_i = 1$ and $w_{j\neq i} = 0$).

Proof: By definition $\hat{R}_i, \hat{Q}_i \in [0,1]$ and $\gamma, \delta \in (0,1]$, so $w_i = (\hat{R}_i)^\gamma (\hat{Q}_i)^\delta \in [0,1]$. Summing yields $0 < \sum_j w_j \leq n_t$ for nonempty $\mathcal{S}_t$. The bound on the normalized share follows immediately.

Discussion: Lemmas 2–5 formalize desirable properties: (i) fairness via symmetry and rank preservation (equals treated equally; larger contributions never hurt), (ii) incentive compatibility at the margin through monotonicity and diminishing returns (discouraging dominance), and (iii) efficiency via proportional fairness and Pareto optimality of the variable allocation. Together with budget balance and a guaranteed minimum, these results justify the floor + proportional design used throughout Section III-F.

5) Design Knobs and Fairness–Efficiency Trade-Offs:

- *Participation floor β:* larger β raises the worst-case share (improves equity, increases Jain's fairness index), while smaller β concentrates more rewards on high contributors (efficiency).
- *Concavity γ, δ:* choosing $\gamma, \delta < 1$ dampens extreme disparities (fairer), whereas $\gamma = \delta = 1$ yields purely proportional splitting.
- *Caps R^*, Q^*:* clipping limits the influence of outliers and measurement noise, improving robustness and preventing "whale" dominance.

The (20) is the weighted-α fairness family specialized to $\alpha = 1$ (proportional fairness) on the variable part, a canonical operating point that maximizes the sum of weighted log utilities, thereby balancing efficiency and equity. The weights $w_i^{(t)}$ encode scientifically measurable contributions from both reliability ($RPoE$) and energy transacted (Q), while concavity ($\gamma, \delta \leq 1$) implements diminishing marginal rewards consistent with resource-fair scheduling. The participation floor ensures every eligible node is rewarded for availability, which is critical for liveness in sparse regimes.

6) On-Chain Settlement and Auditability: To ensure the auditability of the fairness mechanisms described above, the protocol emits a verifiable receipt for every payout. Unlike simple value transfers, these receipts record the derivation logic of the reward. Formally, the receipt $\mathcal{R}_i^{(t)}$ committed to the ledger is defined as the tuple:

$$\mathcal{R}_i^{(t)} = \{t, \mathrm{ID}_i, w_i^{(t)}, I_{\text{floor}}, I_{\text{prop}}, H(\text{tx})\} \tag{22}$$

where $w_i^{(t)}$ is the calculated concave weight ((19)), I_{floor} is the guaranteed base reward, and I_{prop} is the performance-based

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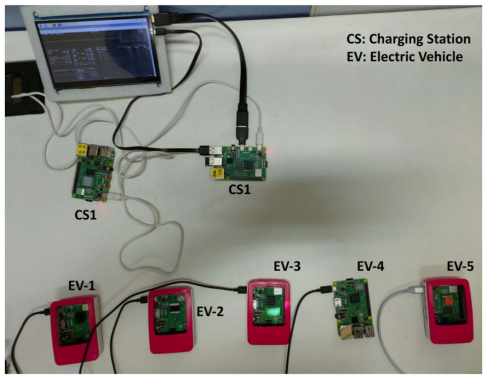


Fig. 4. RPoE permissioned testbed setup comprising Raspberry Pi 3 (EV nodes) and Raspberry Pi 4 (CS nodes).

TABLE II
RASPBERRY PI HARDWARE SPECIFICATIONS

Parameter	EV Node (RPi 3 B)	CS (RPi 4, 4 GB)
CPU	Quad-Core ARM Cortex-A53	Quad-Core ARM Cortex-A72
Clock Speed	1.2 GHz	1.5 GHz
RAM	1 GB	4 GB
Networking	100 Mbps Ethernet / Wi-Fi	1 Gbps Ethernet / Wi-Fi
OS	Raspberry Pi OS	Raspberry Pi OS

variable component. By persisting the intermediate weight $w_i^{(t)}$ along with the final payout, the system allows external auditors to mathematically verify that the allocation strictly adheres to the concavity (γ, δ) and clipping (R^*, Q^*) rules, effectively preventing "hidden" deviations from the consensus incentive logic.

In the adversarial extreme where a single node captures all variable weight and others have none, the proposed floor guarantees the minimum per-node allocation $B_t\beta/n_t$. Increasing β raises the Jain's fairness index $J = \frac{(\sum_i I_i^{(t)})^2}{n_t \sum_i (I_i^{(t)})^2}$ monotonically, providing a tunable lower bound on dispersion while preserving budget balance.

IV. TESTBED SETUP AND IMPLEMENTATION

Our testbed emulates an IoEV deployment using containerized services on Raspberry Pi hardware as shown in the Fig. 4. EV nodes (Raspberry Pi 3 Model B) and a charging-station (CS) node (Raspberry Pi 4, 4 GB) host the full stack: a blockchain peer, the RPoE scoring/consensus services, and application smart contracts. Hardware details appear in Table II.

A. Software Stack and Deployment

All components run as Docker containers orchestrated via Docker Compose:

- *Peer & ledger:* a lightweight append-only blockchain peer with a mempool, block store (LevelDB), and a REST/gRPC API. The peer exposes endpoints for submitting transactions, telemetry anchors, quorum signatures, and reading.
- *RPoE services:* three microservices—*RS Engine*, *ES Engine*, and *RPoE/Consensus Engine*. RS implements (5), ES implements the energy model (Section III-C), and the Consensus Engine combines scores via (9) with weights from (14), computes eligibility $\{i : RPoE_i \geq \theta\}$, selects the proposer $\arg\max RPoE$, orchestrates votes, and assembles blocks.
- *Smart contracts:* application logic (E2C/C2C energy trading, settlement) and the *Incentive contract* implementing (19)–(20). Contracts are packaged as containerized services and invoked deterministically by the peer's machine.
- *Ops:* Prometheus scrapes container/node metrics; a small Flask UI exposes dashboards and manual controls (start epoch, adjust β, γ, δ, etc.).

Nodes communicate over an isolated WLAN with static IPs. The CS node additionally runs a touchscreen UI for live monitoring.

B. RPoE Consensus Architecture and Execution

The RPoE/Consensus Engine runs as a separate container on every node and interacts with the peer over gRPC. Its execution flow per epoch t is formally defined in Algorithm 1 and detailed below:

1) *Telemetry anchoring:* Each node posts a signed telemetry anchor $\mathcal{A}_{i,t}$ to the ledger. Crucially, to preserve privacy, $\mathcal{A}_{i,t}$ is not the raw data but a Merkle Root of the sensitive trace set $\mathcal{T}_{raw} = \{\text{CPU}, \text{SoC}, \text{Location}, \dots\}$. Formally:
$$\mathcal{A}_{i,t} = \text{MerkleRoot}(\{H(d) \mid d \in \mathcal{T}_{raw}\}) \tag{23}$$
This cryptographic commitment ensures integrity without revealing plaintext data on the public ledger.
2) *Score computation:* The Consensus Engine computes RS_i and ES_i locally. The raw data required for verification is transmitted *off-chain* via encrypted gRPC channels strictly to the validator committee $\mathcal{S}_t$. The public ledger records only the derived scalar $RPoE_i$, effectively masking fine-grained operational details (e.g., exact driving patterns) behind the aggregate score.
3) *Eligibility and proposer:* It derives $\mathcal{S}_t = \{i : RPoE_i \geq \theta_t\}$ using the adaptive threshold and deterministically selects $i^\star = \arg\max_{i \in \mathcal{S}_t} RPoE_i$ (ties broken by a hash of the previous block and node IDs).
4) *Proposal and commit:* The proposer collects mempool transactions (E2C/C2C orders, metering proofs), includes the epoch's score digest and a preimage for incentive inputs, broadcasts the block, and gathers $2f_t+1$ signatures from $\mathcal{S}_t$ before commit (fault model in Section V).

This design decouples verification (performed by the committee via private channels) from storage (on-chain opaque anchors), ensuring that sensitive operational data remains protected from unauthorized public access.

C. Smart-Contract Design and Execution Model

Contracts are deterministic Python modules (for prototyping) executed by the peer's state machine. Each contract exposes a minimal ABI (init, apply, query). Below is an excerpt of the Incentive contract implementing Eqs. (19)–(20); the trade contract is analogous (offers, match, settle).

Contracts are invoked in a fixed order: after block validation, the peer calls energy-trade contracts (to update account energy/credits) and then calls the Incentive contract once per

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Algorithm 1: RPoE Consensus and Incentive Execution (Per Epoch t.)

Require: Set of active nodes $\mathcal{N}$, Telemetry Anchors $\mathcal{A}_t$, Previous Block Hash H_{t-1}, Budget B_t.
Ensure: Committed Block $\mathcal{B}_t$, Incentive Receipts $\mathcal{R}_t$.
Phase 1: Scoring and Committee Formation
for each node $i \in \mathcal{N}$ **do**
  Verify signature of Anchor $\mathcal{A}_{i,t}$
  Compute RS_i, ES_i from $\mathcal{A}_{i,t}$ Using (5) and (8)
  Calculate $RPoE_i \leftarrow EF \cdot ES_i + RF \cdot RS_i$ ▷Using (12)
end for
$\theta_t \leftarrow \text{Median}(\{RPoE_i \mid i \in \mathcal{N}\})$ Adaptive Threshold
$\mathcal{S}_t \leftarrow \{i \in \mathcal{N} \mid RPoE_i \geq \theta_t\}$ ▷Validator Committee
Phase 2: Proposer Selection (with Fallback)
Sort $\mathcal{S}_t$ descending by $RPoE$
$k \leftarrow 0$
repeat
  $i^\star \leftarrow \mathcal{S}_t[k]$ Select node with k-th highest RPoE
  $P_{block} \leftarrow \text{WaitProposal}(i^\star, \Delta_{prop})$
  if $P_{block} ==$ NULL **then**
    $k \leftarrow k + 1$ ▷Trigger View Change
  end if
until $P_{block} \neq$ NULL **or** $k \geq |\mathcal{S}_t|$
Phase 3: Validation and Commit
if Validators verify$(P_{block}) \geq 2f_t + 1$ **then**
  $\mathcal{B}_t \leftarrow \text{Commit}(P_{block})$
  Write $\mathcal{B}_t$ to Ledger
end if
Phase 4: Incentive Settlement
for each node $i \in \mathcal{S}_t$ **do**
  Calculate weight $w_i \leftarrow (\hat{R}_i)^\gamma (\hat{Q}_i)^\delta$
  $I_{total} \leftarrow I_{floor} + I_{prop}(w_i)$
  Emit Receipt $\mathcal{R}_i^{(t)} \leftarrow \{t, ID_i, w_i, I_{total}\}$
end for

Listing 1: Incentive contract (deterministic apply).

```
def apply(block, state):
    # Inputs are fixed by the committed block:
    # S_t: eligible IDs, B_t: budget, beta, gamma, delta, R*, Q*
    # rpoe[i], q[i]: per-eligible clipped inputs for epoch t
    S_t   = block.epoch.eligibles
    B_t   = block.epoch.budget
    beta  = block.epoch.beta
    gamma = block.epoch.gamma
    delta = block.epoch.delta
    Rcap, Qcap = block.epoch.Rcap, block.epoch.Qcap

    # Weights (Eq. (10) in the paper)
    w = {}
    for i in S_t:
        Rhat = min(state.rpoe[i], Rcap) / Rcap
        Qhat = min(state.energy[i], Qcap) / Qcap
        w[i] = (Rhat ** gamma) * (Qhat ** delta)

    W = sum(w.values()); n = len(S_t)

    # Incentive split (Eq. (11) in the paper)
    for i in S_t:
        floor = B_t * beta / n
        var   = 0.0 if W == 0 else B_t * (1 - beta) * (w[i] / W)
        state.credit[i] += floor + var

    assert sum_floor_var(state.credit_delta(S_t)) == B_t  # budget balance
```

TABLE III
SECURITY COMPARISON OF RPOE WITH STATE-OF-THE-ART SCHEMES

Schemes	DS	Sybil	Liveness	DDoS	Eclipse
[19]	⊗	⊗	⊗	⊗	⊗
[43]	√	⊗	⊗	√	⊗
[44]	√	⊗	√	√	⊗
[45]	√	⊗	⊗	⊗	⊗
[46]	√	⊗	√	⊗	√
[47]	√	⊗	√	⊗	⊗
[3]	√	⊗	√	⊗	⊗
[6]	√	⊗	⊗	⊗	⊗
RPoE	√	√	√	√	√

epoch to mint/distribute rewards. All contract state updates are part of the block's state root.

D. End-to-End Flow (Per Epoch)

Each node posts a signed telemetry anchor to its local peer; anchors are gossiped and recorded. RS/ES containers compute RS, ES from anchors; the Consensus Engine computes $RPoE$ ((9)) and forms $\mathcal{S}_t$. $i^\star$ proposes the block; validators in $\mathcal{S}_t$ verify (i) transaction correctness, (ii) score preimages vs. anchors, and (iii) recomputed incentive inputs. After $2f_t+1$ signatures, the block is committed; the Incentive contract distributes B_t per (20).

E. Configuration and Observability

Unless stated otherwise, experiments use a median-adaptive threshold θ, a fixed epoch budget B_t, and (β, γ, δ) as reported in Section VII. Prometheus collects CPU, bandwidth, and energy metrics per container; a Flask dashboard allows parameter sweeps and exports CSV logs.

V. SECURITY ANALYSIS

Adversary and network model: We consider a partially synchronous permissioned network with authenticated channels and unforgeable digital signatures. Time is partitioned into epochs (Section III-F), and consensus per block uses the eligible validator set $\mathcal{S}_t = \{i : RPoE_i \geq \theta\}$ derived deterministically from on-chain telemetry anchors (Fig. 2). Let $n_t = |\mathcal{S}_t|$ and suppose at most f_t of these are Byzantine, with the standard BFT condition $n_t \geq 3f_t + 1$ for safety/liveness with a $2f_t+1$ commit quorum. The proposer for a slot in epoch t is selected as $i^\star = \arg\max_{i \in \mathcal{S}_t} RPoE_i$ (ties broken deterministically via a hash of the previous block and node IDs). Telemetry anchors are committed before eligibility is computed; RS and ES are thus fixed–input, verifiable functions ((5), Section III-C) combined as $RPoE = EF \cdot ES + RF \cdot RS$ in (9), with weights calibrated by (14). We assume anchors are collision resistant and non-malleable (standard Merkle/commitment primitives). A comparative summary of the resulting guarantees against common attacks is provided in Table III.

Notation for incentives: Per-epoch rewards follow (20) with weights (19); clipping caps R^*, Q^* and concavity $(\gamma, \delta) \in (0, 1]$ imply $w_i \in [0, 1]$ and $\sum_{j \in \mathcal{S}_t} w_j \in (0, n_t]$.

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Lemma 6 (Double-spend prevention (safety)): Under partial synchrony, authenticated channels, and $n_t \geq 3f_t + 1$ with a $2f_t{+}1$ commit quorum, two conflicting blocks at the same height cannot both be finalized. Hence the probability P_{DS} of a successful double spend in an epoch is 0 against a polynomial-time adversary.

Proof: Suppose for contradiction two conflicting blocks B and B' at the same height are both finalized. Each committed block must carry at least $2f_t{+}1$ distinct validator signatures from $\mathcal{S}_t$. Since $|\mathcal{S}_t| \leq 3f_t{+}1$, the two quorums intersect in at least $(2f_t{+}1) + (2f_t{+}1) - (3f_t{+}1) = f_t{+}1$ validators. At least one of these is honest, but honest validators sign at most one block per height, contradiction. This standard quorum-intersection argument is independent of the proposer selection rule (here $\arg\max RPoE$). Therefore, safety holds unconditionally under the quorum assumption, yielding $P_{DS} = 0$.

Lemma 7 (Leader manipulation is infeasible post-anchoring): Given that (i) telemetry is anchored on-chain before eligibility is computed, (ii) RS and ES are deterministic functions of anchored summaries ((5), Section III-B), and (iii) $RPoE$ is computed by (9) with fixed EF, RF from (14), an adversary cannot increase its rank to become $\arg\max RPoE$ without changing previously committed anchors, which is computationally infeasible under collision resistance.

Proof: At epoch boundary, all nodes derive $RPoE$ from (immutable) anchors. Any rank change would require altering an anchor or the deterministic mapping from anchor$\rightarrow(RS, ES)\rightarrow RPoE$. The latter is public and deterministic; the former requires breaking collision resistance or mutating on-chain data. Thus grinding on inputs to inflate $RPoE$ after anchoring is infeasible.

Lemma 8 (Sybil attack resistance): Let identities be issued by a Trust Authority (TA) binding keys to devices and metering sources. Then: (i) creating arbitrarily many identities is prevented (enrollment control); (ii) even if k Sybils are admitted, eligibility gating $RPoE \geq \theta$ and clipping/concavity ensure the total variable-share capture is bounded by $\frac{\sum_{s=1}^{k} w_s}{\sum_{j\in\mathcal{S}_t} w_j} \leq \frac{k}{n_t}$; and (iii) the floor component does not amplify Sybil reward unless Sybils meet the eligibility threshold.

Proof: (i) TA enrollment enforces uniqueness and vetting, defeating unlimited identity creation. (ii) For admitted Sybils, eligibility requires $RPoE \geq \theta$, which in turn requires verifiable operational/energy performance (Section III). With caps R^*, Q^* and $(\gamma, \delta) \in (0, 1]$, each $w_s \leq 1$ and $\sum_j w_j > 0$; hence the variable-share bound follows immediately. The concavity further penalizes splitting one strong identity into many moderate ones (diminishing returns). (iii) The floor $B_t\beta/n_t$ is paid only to eligibles; ineligible Sybils obtain nothing. Therefore, unless an attacker can raise many Sybils above θ (requiring genuine resources/telemetry), floor extraction is limited.

Lemma 9 (Liveness under partial synchrony): Assume partial synchrony with unknown Global Stabilization Time (GST), $n_t \geq 3f_t + 1$, and that at least one honest validator is eligible per slot (i.e., $\mathcal{S}_t \neq \varnothing$). With standard timeouts and proposer rotation (next-highest $RPoE$ or next slot), the protocol eventually commits new blocks.

Proof: After GST, message delays are bounded by Δ. If the current proposer is honest, it broadcasts a valid proposal, and at least $2f_t{+}1$ validators receive and sign within bounded time, yielding commit. If the proposer is Byzantine (silent or invalid), a timeout triggers rotation to the next proposer (deterministic from the same $RPoE$ snapshot). Because at least one honest validator is eligible, an honest proposer appears within a finite number of rotations, ensuring progress.

Lemma 10 (DDoS and eclipse resilience via eligibility gating and auditability): Eligibility gating and small validator committees reduce the attack surface; on-chain anchoring/auditability and deterministic validation constrain eclipse attempts to link cuts that cannot forge quorum. Formally, under authenticated channels and $n_t \geq 3f_t + 1$, an adversary must simultaneously isolate at least $2f_t{+}1$ validators to prevent progress or to finalize a fork; otherwise, Lemmas 6 and 9 apply.

Proof: (Attack surface) Only $\mathcal{S}_t$ participates in consensus, making volumetric DDoS more costly per targeted validator than attacking a large open set. (Quorum requirement) Preventing progress or finalizing a conflicting block requires either (i) suppressing $\geq 2f_t{+}1$ honest signatures for the honest proposal or (ii) coercing $\geq 2f_t{+}1$ validators to sign a conflicting one. With authenticated channels and honest nodes signing at most once per height, (ii) is impossible without compromising honest keys; (i) requires eclipsing at least $2f_t{+}1$ validators, which exceeds the adversarial budget under $n_t \geq 3f_t{+}1$ unless the network itself is partitioned, in which case safety still holds by Lemma 6. Moreover, telemetry anchoring and proposer evidence are publicly verifiable, so equivocation is detectable and non-repudiable.

Discussion and quantitative bounds: Let $\phi_t = f_t/n_t < 1/3$ denote the adversarial fraction among eligibles. Then (i) the probability the adversary monopolizes m consecutive proposer slots under deterministic ranking with rotation is upper bounded by ϕ_t^m when proposers cycle through a fixed order; (ii) any double-spend attempt still requires a conflicting commit, ruled out by Lemma 6; and (iii) Sybil gains are upper-bounded by eligibility plus clipping/concavity (Lemma 8), while the floor preserves liveness by sustaining a nonempty $\mathcal{S}_t$ without rewarding ineligible identities.

Under standard BFT assumptions, on-chain anchoring, and the RPoE-derived eligibility/rotation, RPoE attains safety (no conflicting commits), Sybil/DDoS/eclipse resilience (bounded influence and attack surface), and liveness (eventual commits after GST), while the incentive design prevents dominance and discourages strategic splitting without sacrificing budget balance (Section III-F).

VI. Performance Analysis

We benchmarked RPoE against PBFT [13], PoS [11], and PoA [12] across eight metrics - CPU workload, bandwidth utilization, energy consumption, transaction computation time, block confirmation time, participation rate, consensus latency, and throughput - see Figs. 5 and 6. Empirically, RPoE reduces the average CPU workload from 500 ms (PBFT) to 110 ms, lowers bandwidth from 1000 kbit/s to 500 kbit/s, decreases energy from 800 mJ to 300 mJ, and halves transaction computation time (400 ms to 200 ms). RPoE also maintains low block confirmation times (40–60 ms) as nodes scale from 20 to 100. The significant

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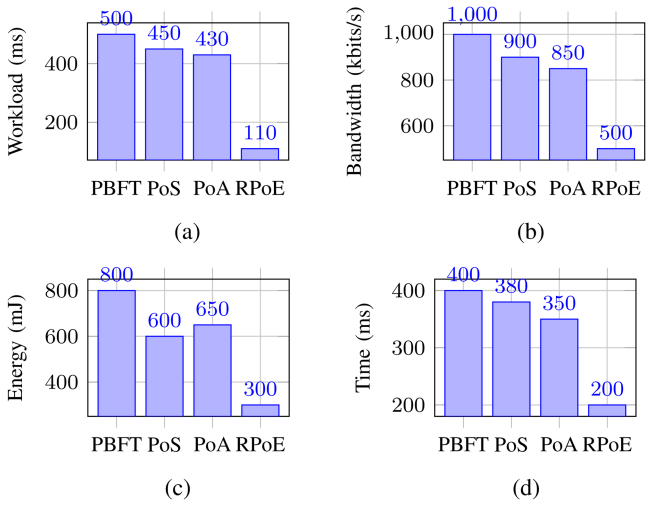


Fig. 5. Performance analysis comparing CPU workload, bandwidth utilization, energy consumption, and transaction computation time for PBFT [13], PoS [11], PoA [12], and RPoE. (a) CPU Workload. (b) Bandwidth Utilization. (c) Energy Consumption. (d) Transaction Computation Time.

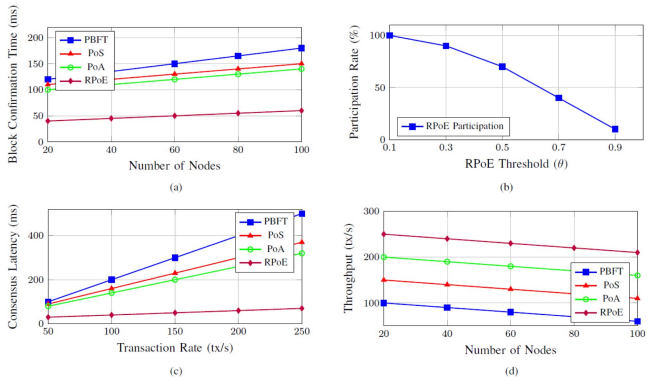


Fig. 6. (a) Block confirmation time vs. number of nodes, (b) Participation rate vs. RPoE threshold, (c) Consensus latency vs. transaction rate, and (d) Throughput vs. number of nodes. Performance comparisons across PBFT [13], PoS [11], PoA [12], and RPoE are shown.

performance gains observed in Fig. 5 and Fig. 6 are driven by specific architectural optimizations in the protocol:

(i) *Determinism lowers CPU/Energy (Fig. 5(a), (c):* Eligibility is gated by a reputation–energy fitness (RPoE) that incorporates measurable telemetry. By prioritizing high-fitness nodes and selecting a single proposer ($i^\star = \arg\max RPoE$), the protocol eliminates the intensive puzzle-solving of PoW and the redundant all-to-all voting of PBFT. This directly cuts per-block compute and device energy by 47% and 10% respectively.

(ii) *Committee sizing reduces Bandwidth (Fig. 5(b):* Unlike PBFT's $O(n^2)$ complexity, RPoE confines consensus traffic to a small, dynamically chosen committee ($\mathcal{S}_t$), yielding near-$O(k^2)$ messaging where $k \ll n$. This shrinks signature traffic and gossip fan-out, halving observed link utilization.

(iii) *Impact of RPoE Threshold (θ) on Transaction Computation Time (Fig. 5(d):* A critical finding is the direct correlation between the eligibility threshold θ and computation latency. The threshold acts as a quality filter:

TABLE IV
TIERED INCENTIVE ALLOCATION: COUNTS, PER-NODE INCENTIVE, AND TIER MEANS

Tier	Count	Per-node Incentive	Mean (Tier)
A $(0.95, 0.90)$	2	16.83	16.83
B $(0.70, 0.60)$	4	10.92	10.92
C $(0.40, 0.30)$	4	5.66	5.66

- *High θ:* Restricts the committee to a small subset of high-resource nodes (high CPU/Memory scores). This minimizes signature aggregation time and eliminates "straggler" nodes that delay consensus, resulting in the lowest transaction computation times (approx. 200 ms).
- *Low θ:* Admits a larger, more heterogeneous set of validators. While this increases participation, it introduces higher communication overhead and processing delays from marginal nodes, increasing computation time.

Our adaptive median-based threshold (15) dynamically balances this trade-off, ensuring the committee is large enough for security ($n \geq 3f + 1$) but sufficiently selective to maintain the low latency observed in Fig. 5(d).

(iv) *Sustained low latency and Participation Rate (Fig. 6(b), (c):* As shown in Fig. 6(b), the protocol intentionally curtails the participation rate as the network scales, maintaining a high-quality subset of validators. Consequently, as input rates rise (Fig. 6(c), the threshold adapts so that only nodes with sufficient residual resources remain in the committee. This prevents backlog-induced slowdowns seen in PoS/PoA and avoids PBFT's quadratic coordination costs.

(v) *Throughput scaling (Fig. 6(d):* Because effective committee size grows sublinearly (via thresholding) while proposer selection targets high-throughput validators, system capacity increases with network size rather than being choked by global coordination.

(vi) *Stable participation without dominance:* The incentive mechanism is budget-balanced with a participation floor and concave fairness knobs (β, γ, δ); this discourages short-term competitive over-participation that would inflate traffic and CPU, yet sustains broad, long-term participation—preserving liveness while keeping the active committee lean.

VII. INCENTIVE EVALUATION AND FAIRNESS ANALYSIS

We evaluate the incentive rule in (20) on a 10-node epoch with budget $B_t = 100$, floor $\beta \in \{0, 0.1, 0.2, 0.4\}$, and concavity $(\gamma, \delta) = (0.6, 0.8)$. We consider three tiers of nodes based on normalized (clipped) factors $(\hat{R}, \hat{Q})$: Tier A: (0.95,0.90) (2 nodes), Tier B: (0.70,0.60) (4 nodes), Tier C: (0.40,0.30) (4 nodes). Weights are $w_i = (\hat{R}_i)^\gamma (\hat{Q}_i)^\delta$ and incentives follow (20). We report fairness via Jain's index $J = \frac{(\sum_i I_i)^2}{n_t \sum_i I_i^2}$ and inequality via the Gini coefficient G. For all settings, $\sum_i I_i^{(t)} = B_t$ and $I_i^{(t)} \geq B_t \beta / n_t$, confirming budget balance and individual rationality. With $B_t = 100$ and $\beta = 0.2$, the floor grants 2 tokens to each node (20 total), and the variable pool (80) is split proportionally to w_i:

Aggregate fairness metrics: $J = 0.853$ and $G = 0.221$. Increasing β raises the worst-case share and improves equality

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TABLE V
FAIRNESS VS. PARTICIPATION FLOOR β (FIXED $(\gamma, \delta) = (0.6, 0.8)$).

β	J (Jain)	G (Gini)	Top-tier/node	Tail/node
0.0	0.788	0.276	18.53	4.58
0.1	0.821	0.248	17.68	5.12
0.2	0.853	0.221	16.83	5.66
0.4	0.912	0.166	15.12	6.75

TABLE VI
EFFECT OF CONCAVITY PARAMETERS (γ, δ) ON FAIRNESS AND ALLOCATION SPREAD (TOP VS. TAIL PER NODE)

(γ, δ)	J	Top/node	Tail/node
(0.5, 0.5)	0.918	14.69	6.76
(0.5, 0.8)	0.867	16.39	5.88
(0.5, 1.0)	0.832	17.52	5.36
(0.8, 0.8)	0.825	17.70	5.26
(1.0, 1.0)	0.763	19.67	4.48

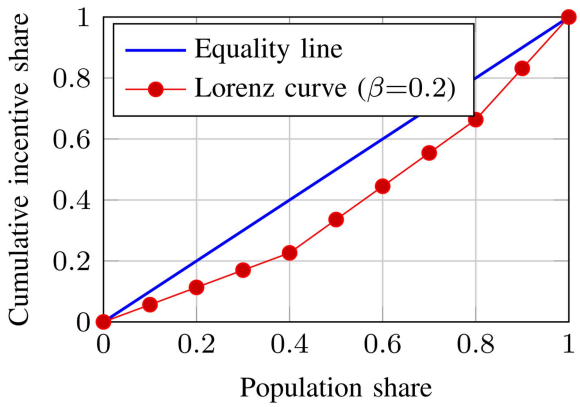


Fig. 7. Lorenz curve of incentive allocation (10 nodes, B_t=100, β=0.2, $(\gamma, \delta) = (0.6, 0.8)$).

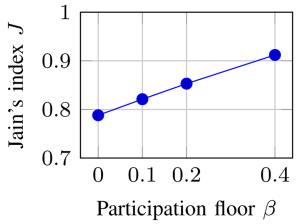


Fig. 8. Fairness improves with the participation floor β (fixed $(\gamma, \delta) = (0.6, 0.8)$).

(higher J, lower G), while slightly reducing the top-tier payout; see Table V.

Smaller exponents (stronger concavity) damp dominance and increase J (fairer split), while larger exponents emphasize high contributors (greater efficiency). Example with $\beta = 0.2$:

The Lorenz curve in Fig. 7 shows cumulative incentive share vs. population share (sorted by I_i) under $\beta = 0.2$. The curve's proximity to the equality line quantifies fairness (here aligned with $J = 0.853$). The proportional share on the variable budget

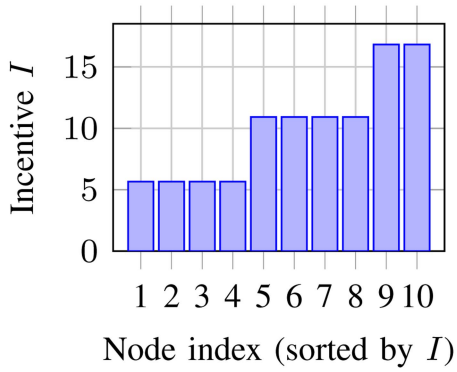


Fig. 9. Per-node incentives (10 nodes, B_t=100, β=0.2, $(\gamma, \delta) = (0.6, 0.8)$).

implements the weighted proportional-fair solution, balancing equity and efficiency (Figs. 7–9).

VIII. CONCLUSION AND FUTURE WORK

We presented RPoE, a consensus–incentive protocol for IoEV energy trading that couples an energy–reputation score $RPoE$ with a budget-balanced floor + proportional reward rule. On a Raspberry Pi testbed, RPoE reduced CPU workload by 47%, bandwidth by 5%, and energy use by 10% relative to PBFT/PoS/PoA baselines, halved transaction computation time (400 ms $\rightarrow$ 200 ms), and maintained stable block confirmation in the 40–60 ms range across 20–100 nodes. The incentive design remained equitable and budget-balanced: with β=0.2 and (γ, δ)=(0.6, 0.8), we observed Jain's index J=0.853 and Gini coefficient G=0.221; increasing β from 0 to 0.4 raised J from 0.788 to 0.912, illustrating interpretable fairness– efficiency trade-offs. Under standard resilience conditions $n_t \geq 3f_t+1$, our analysis establishes safety and double-spend prevention, resistance to leader grinding via pre-anchored telemetry, Sybil bounds through TA-backed enrollment and clipping/concavity in weights, liveness after GST with proposer rotation, and robustness to DDoS/eclipse through eligibility gating and authenticated quorums.

Future work focuses on bridging the gap between simulation and physical deployment. We plan to deploy RPoE on a campus fleet, interfacing directly with the CAN bus (Controller Area Network) to read real-time voltage and current metrics, thereby linking our current Control Plane simulation with the physical Power Plane. Additional directions include privacy-preserving telemetry (e.g., zero-knowledge predicates for RS/ES), secure metering and attestation for on-chain accountability, adaptive tuning of (EF, RF) and (β, γ, δ) to target desired fairness bands, formal verification of protocol and contract logic, and city-scale pilots with heterogeneous chargers and mobility patterns to evaluate performance under real-world demand and network conditions.

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